

SAMPLE PAPER (synthetic, for testing uploads)

B.Tech Sem III Theory Examination 2023

Data Structures and Algorithms II

Time: 3 Hours Total Marks: 70

SECTION A

1. Attempt all questions in brief.

(a) Define a threaded binary tree.

(b) What is the balance factor of a node in an AVL tree?

(c) Differentiate between BFS and DFS.

(d) State the principle of optimality in dynamic programming.

(e) What is a B-tree of order m ?

SECTION B

2. Insert 50, 20, 60, 10, 8, 15, 32, 46, 11, 48 into an AVL tree. Show every rotation. (10 marks)

3. Find the minimum spanning tree of the given graph using Kruskal's algorithm. Compare it with Prim's algorithm. (10 marks)

4. Apply Dijkstra's algorithm to find the single source shortest path from vertex A. (10 marks)

5. Solve the 0/1 knapsack problem for $n = 4$, $W = 8$ using dynamic programming. (10 marks)

6. Find the longest common subsequence of ABCBDAB and BDCABA. (10 marks)

SECTION C

7. Explain heap sort. Build a max heap from 4, 10, 3, 5, 1. (10 marks)

8. Solve the 4 queens problem using backtracking. Draw the state space tree. (10 marks)